



Design Overview

Sambridda Ranabhat
Commercial Water Management

Introduction

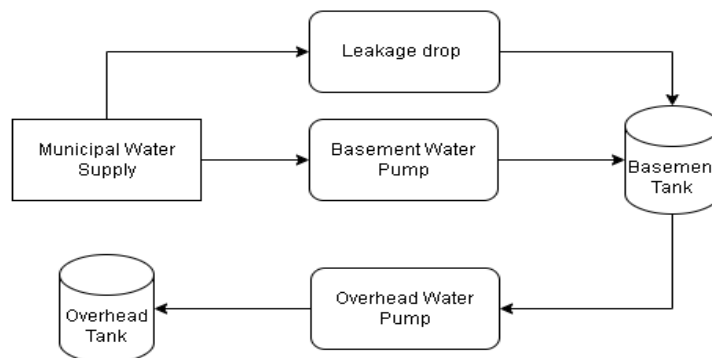
Residential water management in many regions remains a fragmented process, characterized by manual monitoring and open-loop control. Homeowners typically rely on visual inspection or mechanical float switches to manage storage levels and pump operations. However, these traditional methods are prone to human error and mechanical fatigue, often resulting in catastrophic pump failure due to dry-running or significant water wastage through overflows.

This project addresses these inefficiencies by developing an intelligent, distributed Automation system designed for residential water resource management. By transitioning from a manual approach to a semi-automated, closed-loop architecture, the system integrates embedded controllers with real-time sensor networks. The primary focus of this implementation is to optimize operational efficiency, provide robust protection for electromechanical hardware, and enhance overall user safety through autonomous fault detection.

The current architecture serves as a Functional Prototype. The next iteration involves migrating the control logic to an Industrial PLC (e.g., Siemens S7-1200). This would replace the 433MHz link with RS-485/Modbus for electrical noise immunity and utilize industrial 24V DC logic for integration with standard motor control centers (MCCs).

Technical Specification

The domestic water architecture is configured as a dual-stage hydrostatic system designed to bridge the gap between intermittent municipal supply and consistent residential demand. The primary interface with the municipal utility is a regulated, metered connection that serves as the official point of custody transfer for billing and taxation purposes. This connection feeds into a low-level basement reservoir, which acts as the system's primary buffer. While this reservoir is largely populated during active municipal supply windows, the system also accounts for "passive ingress"—low-velocity leakage that occurs when the supply is active but lacks the pressure for high-volume flow. Though this flow rate is significantly lower than the mechanical discharge rate of the pump, it represents a secondary, albeit slow, recovery method for the basement storage volume.



The transition of fluid from the basement to the terminal storage is managed by a vertical lift assembly. This pump is responsible for overcoming the static head required to elevate water to the overhead distribution tank. Functioning as the system's "header tank," the overhead unit utilizes gravitational potential energy to provide pressurized flow to all internal fixtures. In this architecture, the overhead tank is defined as the critical path component; because it provides the necessary pressure for distribution, any depletion of this volume to "dead storage" levels results in a total loss of hydraulic service to the household. Consequently, the automation of the lift assembly is vital to maintaining the equilibrium between the basement supply and the overhead distribution requirements.

Assumptions & Constraints

Assumptions

- **Constant Volumetric Geometry:** It is assumed that both the Primary Reservoir and Secondary Tank are fixed parameters where the vertical distance measured by the ultrasonic sensor correlates linearly to water volume.
- **Constant Pump Head & Flow Rate:** We assume the pump operates at a consistent flow rate given a stable voltage. This justifies using Time-Based Safety Cutoffs as a secondary fail-safe to the current sensor.
- **Non-Turbulent Surface Conditions:** It is assumed that the inflow rate into the tanks does not create enough surface agitation to cause "noise" or false readings in the JSN-SR04T ultrasonic data.
- **Power Grid Recovery Logic:** We assume that upon power restoration after a "load-shedding" event, the system should enter a Safe-Boot state (60-second delay) rather than immediately triggering high-current inductive loads (the pump).

Constraints

- **Legal/Regulatory (Non-Invasive Sensing):** The municipal supply line is metered and protected by local utility laws. The system must be strictly "Post-Meter." No modification or tampering with the municipal inlet pipe is permitted, which precludes the use of inline flow meters at the source.
- **Physical/Structural (Fixed Plumbing):** The building's existing plumbing is encased in concrete or finished walls. Re-plumbing for flow sensors or internal float switches is prohibited due to high labor costs and structural risk. The solution must be Retrofit-Ready, utilizing external or non-contact sensing (Ultrasonic).
- **Economic (Budgetary Cap):** As a student/independent project, the hardware budget is strictly limited. This dictated the use of 433MHz RF modules (cost-effective) over LoRa, and the choice of ESP-series microcontrollers over industrial PLCs.
- **Environmental (Power Instability):** The operating environment is subject to frequent and unpredictable power outages. The system must include non-volatile memory (EEPROM/LittleFS) to "remember" its state (e.g., was the pump on before the power cut?) to prevent logic loops upon reboot.

- RF Interference: The communication must pass through reinforced concrete floors. This limits the data packet size and requires a high-redundancy "Burst-and-Ack" protocol to overcome signal attenuation.

Objectives

The primary objective of this project is to engineer a high-reliability, closed-loop automation system that transitions the facility from manual oversight to a fully autonomous water management architecture. Beyond simple fluid transfer, the system aims to achieve optimal resource equilibrium by utilizing real-time sensor feedback to prevent dry-run scenarios in the lift assembly and overflow waste in the secondary storage. Furthermore, the project seeks to integrate fault-detection logic to identify anomalies—such as municipal supply failure or mechanical pump fatigue—ensuring that the overhead distribution remains uninterrupted. By digitizing the monitoring process, the system will effectively decouple household water availability from human intervention, creating a "set-and-forget" utility environment that prioritizes system longevity and consistent hydraulic pressure.

Strategic Advantages of System Automation

Transitioning from a manual operational model to an automated framework provides a significant increase in mechanical efficiency and infrastructure preservation. In a manual system, pumps are frequently subjected to "dry-run" conditions or excessive cycling, which accelerates motor wear and leads to premature component failure. An automated controller, however, ensures the pump operates strictly within its optimal performance curve and only when hydraulic conditions are met. Beyond hardware longevity, automation introduces a layer of precision resource management; by eliminating human error, the risk of reservoir overflow is mitigated, directly reducing water waste and preventing potential structural damage to the basement or roof areas. Most importantly, the shift provides operational continuity, as the system can react to municipal supply fluctuations in real-time, ensuring the critical overhead tank is always replenished during supply windows without requiring the physical presence of an operator.

Technical Implementation Framework

Distributed Intelligence & Multi-Processor Topology

The system has been migrated to a heterogeneous multi-processor architecture to better handle concurrent telemetry and actuation. The Overhead Station (OS) retains the ESP8266 for low-power edge sensing, while the Sump Station (SS) and Control Station have been upgraded to the Dual-Core ESP32 platform. This shift enables the use of a Real-Time Operating System (RTOS), allowing the system to handle high-speed tasks—like ACS712 current sampling—on one core, while managing long-range RF communication and HMI updates on the other.

Advanced Hardware Integrity: Shielding & Isolation

To ensure industrial-grade reliability, the system now utilizes a custom-engineered PCB designed with a strict focus on Electromagnetic Compatibility (EMC).

- **EMF Shielding:** Strategic ground planes and shielding traces have been implemented to mitigate interference from high-voltage inductive pump surges.
- **AC/DC Galvanic Isolation:** The high-voltage actuation circuit is physically and electrically isolated from the low-voltage logic traces. This design adheres to rigorous safety standards by maximizing Creepage and Clearance distances, effectively preventing high-voltage arcing and protecting the sensitive ESP32 silicon from catastrophic failure.

Industrial Design & HMI Development

Beyond functional code, the project prioritizes Industrial Design and Human-Machine Interface (HMI) development. The control enclosure is engineered for ingress protection (IP-rated) to withstand the high-humidity environments typical of pump rooms. The HMI serves as the visual bridge between the machine and the user, providing a high-level dashboard of system health, current reservoir volumes, and pump status. By following ergonomic and intuitive design principles, the interface allows for rapid status assessments and simplified manual overrides, ensuring that the technology remains accessible during maintenance or emergency scenarios.

Enhanced HMI & Control Interface

The system features a centralized Human-Machine Interface (HMI) centered around a 2.4-inch TFT LCD panel. This display serves as the primary visual output for a custom-developed Embedded Operating System (OS), which manages the complex state-machines of the automation logic. Navigation and system configuration are handled via a tactile tri-button interface, allowing the operator to toggle between real-time telemetry, historical data logs, and manual configuration menus.

To provide a multi-sensory user experience, the system integrates a frequency-controlled piezo-buzzer. This component serves two distinct roles:

1. **Haptic/Auditory Feedback:** Providing short, low-decibel "pings" to confirm user inputs and menu navigation, ensuring a responsive interaction with the OS.
2. **Emergency Annunciation:** In the event of a critical system fault—such as a dry-run condition or a sensor failure—the buzzer transitions to a high-intensity alarm pattern to immediately alert residents of a loss in hydraulic integrity.

Communication & System Topology

The system utilizes a decentralized star topology managed by a centralized Orchestrator (the Control Station). Unlike traditional linear controllers, this architecture treats each node as an independent intelligent agent, communicating over a robust, low-latency wireless protocol. By migrating the SS and Control Station to the dual-core ESP32 platform, the topology now supports Asynchronous Data Exchange, ensuring that telemetry from the Overhead Station (OS) does not block critical fault-detection routines in the Sump Station (SS).

The communication layer is built upon an RTOS-integrated messaging protocol. By leveraging the ESP32's ability to run a dedicated radio task on Core 0, the system achieves near-deterministic communication intervals. To combat the harsh electromagnetic environment created by high-torque pump motors, the system implements a Software-Defined Filtering (SDF) layer. This ensures that transient noise—mitigated

physically by the EMF-shielded PCB—is further scrubbed at the data level, preventing false triggers or "ghost" commands from interrupting the system state.

The topology is categorized into three distinct operational tiers:

- The OS (Overhead Station): Acts as the Remote Sensing Node. It maintains a low-power state, periodically transmitting high-precision ultrasonic telemetry to the Orchestrator.
- The SS (Sump Station): Functions as the Power Actuation Node. It is the most hardware-intensive site, managing the AC/DC isolated power circuits and performing real-time current analysis via the ACS712 sensor. It operates with local autonomy; if communication with the Master is lost, the SS can execute an emergency "Local Halt" based on its own sensor data.
- The Control Station: Acts as the System Orchestrator and HMI Provider. It aggregates data from both the OS and SS to maintain the global system state. It manages the user-facing Pocket OS on the 2.4" LCD, processing input from the tri-button interface and governing the frequency-modulated buzzer for auditory alerts.

Software Evolution: From Bare-Metal to RTOS

The previous "Barebone OS"—which functioned as a deterministic, non-blocking polling loop—has been deprecated in favor of a Preemptive Multitasking Architecture. By leveraging the ESP32's native RTOS, the system no longer relies on sequential execution. Instead, critical safety tasks (such as Fault-Tolerance & Fail-Safe Logic) are assigned the highest priority levels. This ensures that a "Dry-Run" or "Overload" event triggers an immediate hardware interrupt, bypassing routine level-checks and ensuring a sub-millisecond response time to mechanical anomalies.